# Locks and keys: public-key cryptography



## Cyber security · Level 4 · Lesson 2

# In this lesson you will

- Explain the key-sharing problem and how two keys solve it.
- Explain what a digital signature proves.
- Say what the HTTPS padlock does and doesn’t prove.

# Key ideas

- Public key: share it, anyone can lock
- Private key: keep it, only it opens
- Sign with the private key, check with the public
- Padlock: private, not honest

# Activity: Padlock post

1. Warm-up (3 min): The key problem
2. Lab (6 min): Make a key pair
3. Act it out (5 min): Padlock post
4. Wrap-up (3 min): What the padlock proves

**One padlock with its key and a small box per group of three, or draw them on paper.**

# Think it through

1. Why is it safe to give your public key to everyone?
2. How does a signature stop a fake phone update?
3. Why can a scam site still show a padlock?

# Before you go

- Which key locks, which opens, and which signs?